

Seat No: _____

Enrollment No: _____

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech. Summer 2023 - 24 Examination

Semester: 4

Date: 24/04/2024

Subject Code: 303105210

Time: 02:00 pm to 04:30 pm

Subject Name: Computer Organization and Microprocessor

Total Marks: 60

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1	Objective Type Questions - (Fill in the blanks, one word answer, MCQ-not more than Five in case of MCQ) (All are compulsory) (Each of one mark)	(15)	CO	PO	Bt
	1. Define: Opcode and Operand.				
	2. If S0=0 and S1= 1 then which operation microprocessor has to perform?				
	3. What is the function of ALE?				
	4. Which interrupt has maximum priority?				
	5. Explain usage of STAX instruction.				
	6. How many transistors are used in 8085 microprocessor?				
	7. Define Machine cycle				
	8. In STA 2050H instruction, __ is Opcode and __ is Operand.				
	9. AR Stands for _____				
	10. 8085 has _ address bit. a. 4 bits b. 8 bits c. 32 bits d. 64 bits				
	11. Define Stack and subroutine				
	12. How many T-states are required to execute instruction MOV B,C ?				
	13. MVI C, 20H is a _ byte instruction. a. 1 Byte b. 3 Byte c. 2Byte d. None				
	14. What is MIMD and SIMD?				
	15. Explain INR instruction.				
Q.2	Answer the following questions. (Attempt any three)	(15)			
	A) Draw pin diagram of 8085 and explain any 5 pins in detail				
	B) Explain flag register in detail.				
	C) Draw and explain programmable model of 8085				
	D) Explain MAR and MAD with memory diagram.				
Q.3	A) Draw and explain timing diagram for MVI B, 05H.	(07)			
	B) What is the instruction and how it will be fetch and executed?	(08)			
	OR				
	B) Explain the stack and subroutine with role of PUSH and POP instructions.	(08)			

Q.4	A) Classify Interrupts and explain it with example.	(07)			
	OR				
	A) Explain RIM and SIM instruction in detail	(07)			
	B) Draw memory interfacing of EPROM of 2Kbyte and RAM of 4Kbyte with 8085 using the decoder.	(08)			